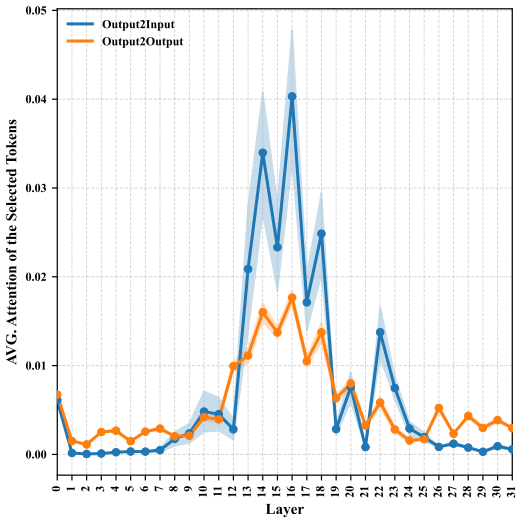
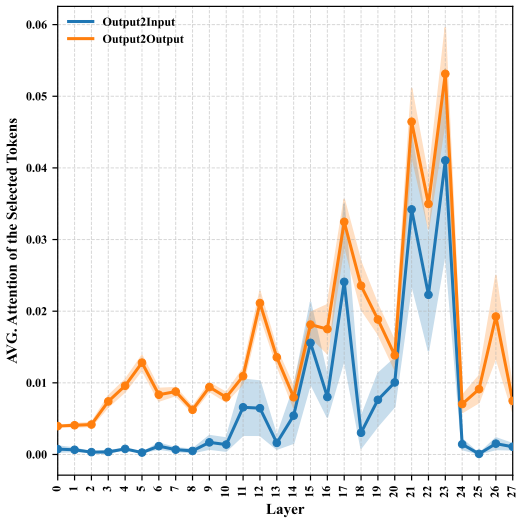


(a) Average attention weights across layers for Llama 3.2 8B. We report attention from selected output tokens to both input and output tokens. Attention to the key partition-related tokens peaks consistently between layers 13 and 18, motivating our choice of this interval for full attention visualization.



(b) Average attention weights across layers for Gemma 3 4B. We report attention from selected output tokens to both input and output tokens (as defined in Fig. 4a). Attention magnitude peaks between layers 21 and 23.

Figure 11: Layer-wise attention analysis for Llama 3.2 8B and Gemma 3 4B. For each model, we visualize the average attention from selected output tokens to salient input tokens (notably the final item and comma of the partition). These trends are used to identify the layer ranges with the strongest partition-relevant attention.

of both intermediate transfer and final summation (the exact prompt template is provided below).

```
I will provide an input text containing fruits
    separated by the '|' delimiter. Your goal
    is to count the items in each section and
    provide a summation.

Input Text:
 apple, apple, apple | apple, apple, apple,
    apple, apple | apple, apple, apple, apple

Task:
1. Identify each partition separated by '|'.
2. Count the number of fruits in each specific
    partition.
3. Sum the counts of all partitions.

You must strictly follow this format, adapting
    the number of parts to the actual input:
part1: <count1>, part2: <count2>, ... , final
    count: <total_count>
```

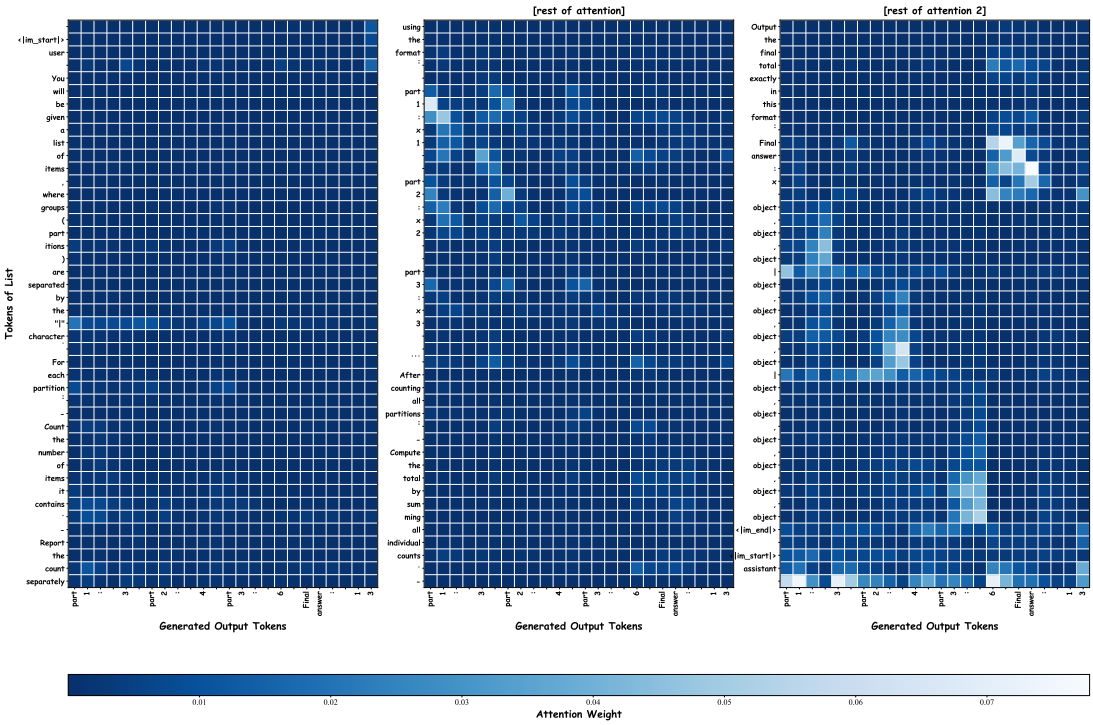


Figure 12: Full attention map for Qwen2.5-7B. The x-axis corresponds to output tokens and the y-axis to input tokens. The token immediately preceding the generated number of parts exhibits the strongest attention toward the final item and trailing comma of each partition matches its own segment.

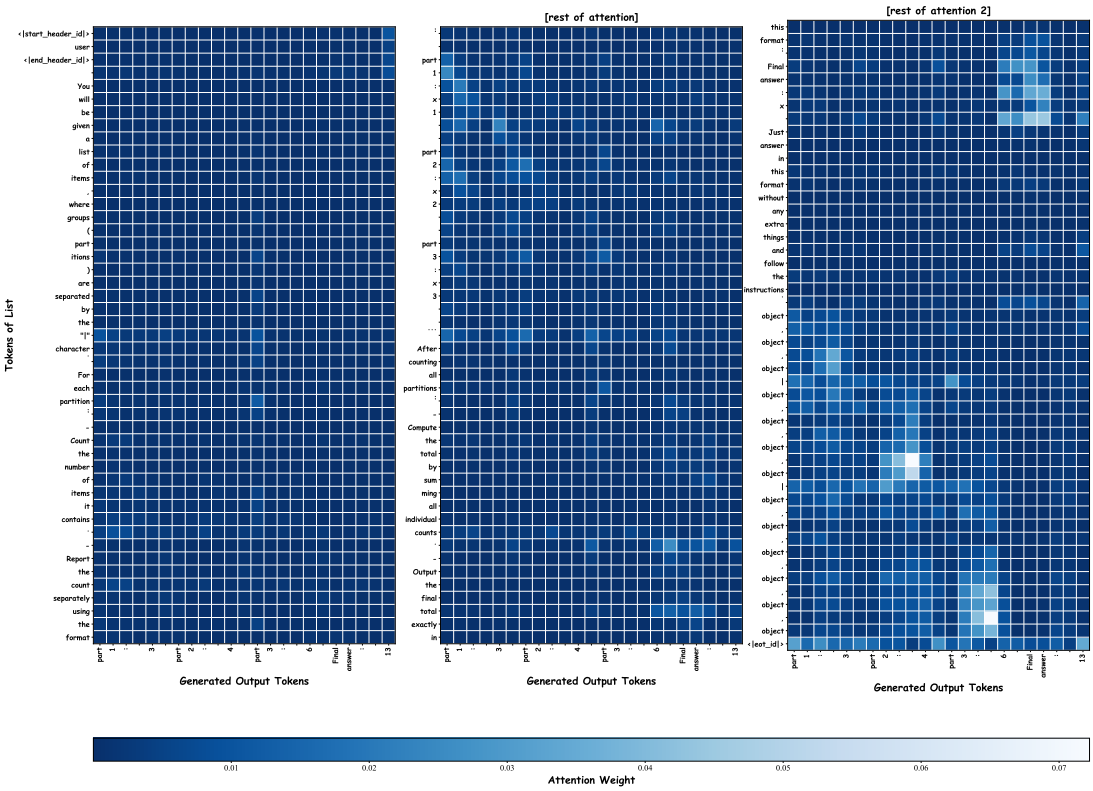


Figure 13: Full attention map for Llama 3.2 8B. The x-axis corresponds to output tokens and the y-axis to input tokens. The token immediately preceding the generated number of parts exhibits the strongest attention toward the final item and trailing comma of each partition matches its own segment.

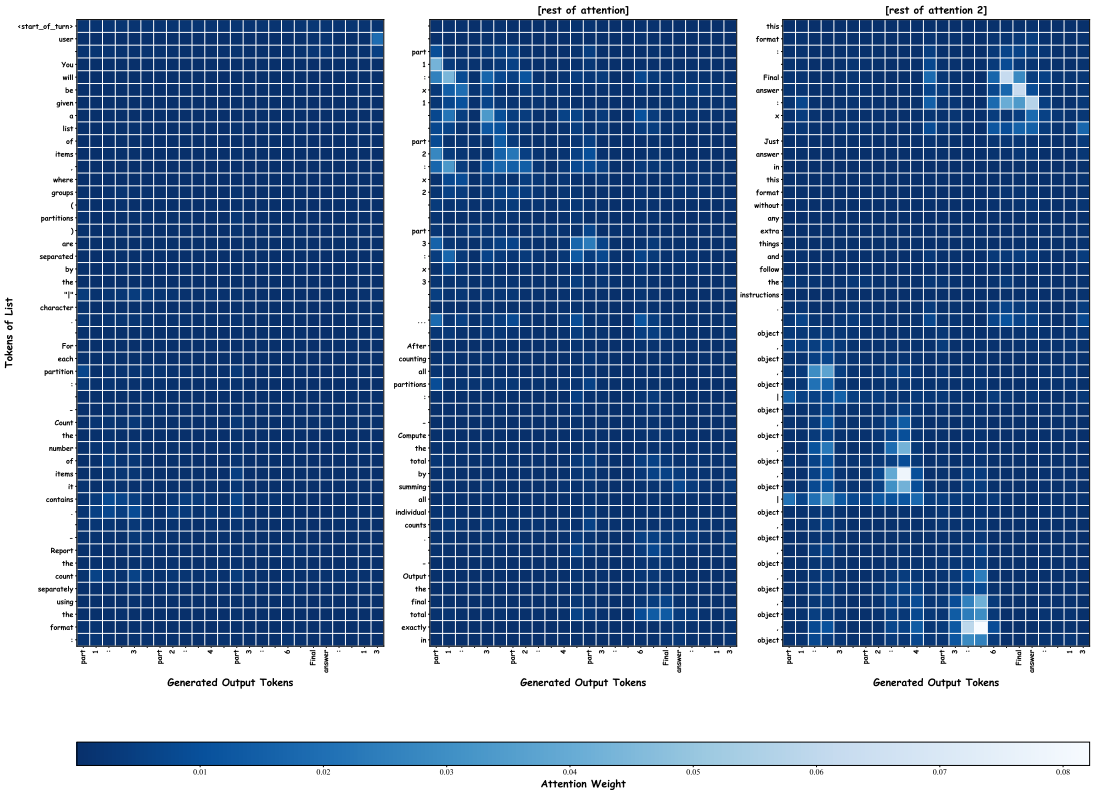
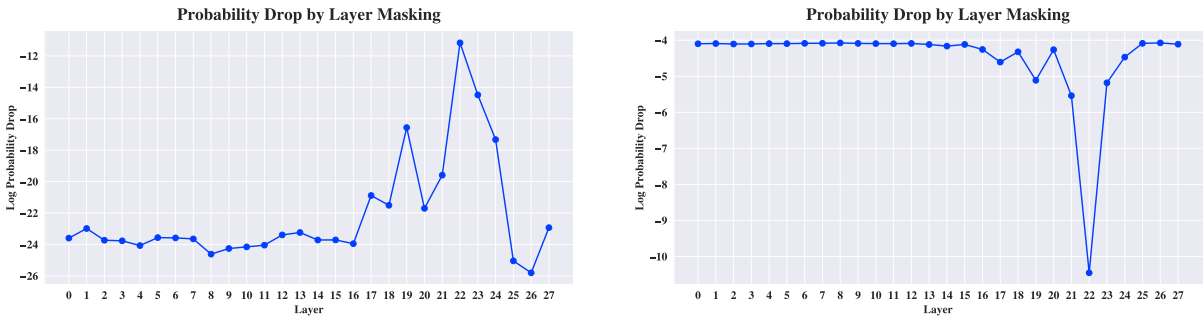


Figure 14: Full attention map for Gemma 3 4B. The x-axis corresponds to output tokens and the y-axis to input tokens. The token immediately preceding the generated partition-level count exhibits the strongest attention toward the final item and trailing comma of each partition matches its own segment.



(a) (b)

Figure 15: Average change in log probability of intermediate counts after layer masking (a) and unmasking (b) of the final item and separator of each partition. In the masking experiment, embeddings from the target layer are zero-ablated. In the unmasking experiment, embeddings of the selected tokens are zero-ablated across all layers, and only the target-layer embeddings are restored from the clean run.